Modern Algorithmic Game Theory

Martin Schmid
Department of Applied Mathematics
Charles University in Prague

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Recap

- In the last two weeks, we started talking about normal-form games, introduced two solution concepts and their properties, and finally described Support Enumeration, an algorithm that finds all possible Nash equilibria in a given game.
- The problematic part about this algorithm is that its runtime is exponential in the number of actions and solving larger games quickly becomes intractable.
- Today, we will first focus on a couple of multi-agent dynamics in non-zero-sum games
- We will then come back to zero-sum games and introduce the first self-play style algorithm – *Fictitious Play*


Unexpected Dynamics in Game Theory

Congestion Games and The Price of Anarchy

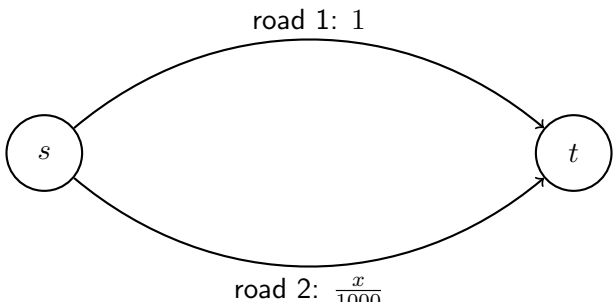


Figure: 1000 drivers travel from s to t, x is the number of drivers on road 2

Congestion Games and The Price of Anarchy

- In a *congestion game*, players choose resources (e.g., roads), and the cost of using a resource depends on how many players use it
- In our example, 1000 drivers travel from s to t and each chooses one of two roads:
  - road 1 always takes 1, no matter how many drivers use it
  - road 2 takes $\frac{x}{1000}$, where x is the number of drivers using it
- Each driver wants to minimize their own delay
- We will compare two outcomes:
  - the *social optimum*, which minimizes the total delay of all drivers
  - the Nash equilibrium, which results from drivers acting selfishly

Congestion Games and The Price of Anarchy

- First, let us compute the minimum possible delay
- Suppose x drivers go to road 2 and $1000 - x$ go to road 1
- Then, the total delay is

$$\frac{x^2}{1000} + (1000 - x)$$

- This expression is minimized when $x = 500$, that is, 500 drivers go to road 2 and the other 500 to road 1
- The total delay is then

$$500 \times \frac{1}{2} \; + \; 500 \times 1 \; = \; 750 \text{ minutes}$$

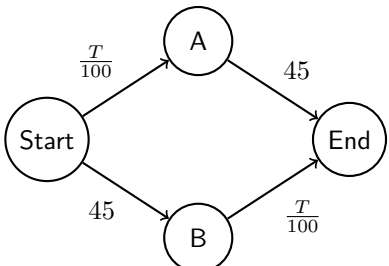
Congestion Games and The Price of Anarchy

- However, drivers on road 1 are not happy – road 2 takes only $\frac{1}{2}$, so each of them has an incentive to switch
- Road 2 never takes longer than road 1, so (essentially) all drivers end up on road 2 in a Nash equilibrium
- The total delay is then $1000 \times 1 = 1000$ minutes, even though no driver can improve by changing roads
- The *Price of Anarchy* (PoA) measures how much selfish behavior costs us

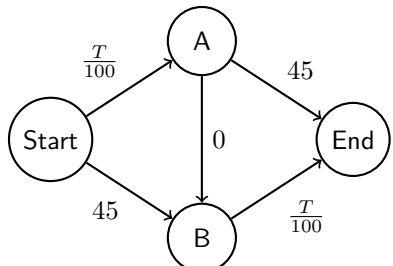
$$PoA = \frac{\text{total cost of the worst Nash equilibrium}}{\text{total cost of the social optimum}} = \frac{1000}{750} = \frac{4}{3}$$

- In fact, $\frac{4}{3}$ is the worst possible PoA in routing games where delays are linear functions of the traffic (Roughgarden and Tardos, 2002)

Braess' Paradox



(a) Original roads/actions



(b) Adding a new road/action

Braess' Paradox

- 4000 drivers travel from Start to End, T denotes the number of drivers on a road
- Original network: two routes, Start $\rightarrow$ A $\rightarrow$ End and Start $\rightarrow$ B $\rightarrow$ End, each consisting of one congested road ($\frac{T}{100}$ minutes) and one fixed road (45 minutes)
- The routes are symmetric, so in a Nash equilibrium, 2000 drivers take each of them and each driver needs

$$\frac{2000}{100} + 45 = 65 \text{ minutes}$$

Braess' Paradox

- New network: we add a road from A to B that takes 0 minutes
- The congested road Start $\to$ A takes at most $\frac{4000}{100} = 40 < 45$ minutes, so it is always better than the fixed road Start $\to$ B; the same holds for B $\to$ End and A $\to$ End
- Hence, taking Start $\to$ A $\to$ B $\to$ End is a dominant strategy, and in the Nash equilibrium, every driver needs

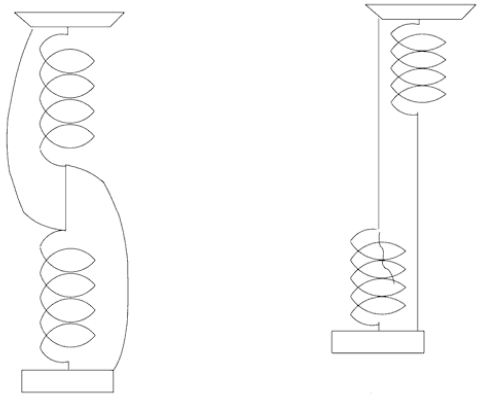
$$\frac{4000}{100} + 0 + \frac{4000}{100} = 80 \text{ minutes}$$

- Adding a road made *everyone* worse off!

Braess' Paradox

- This actually sometimes happens!
  - e.g., traffic in Seoul improved after the Cheonggyecheon highway was demolished in 2003
- Could this happen in a single-agent game?
  - No – a single agent can always ignore the new action, so adding an action can never make it worse off
  - In a multi-agent game, a new action changes what the *other* agents do, and that can hurt everyone
- Are agents rational?
  - Yes – every driver plays a best response (even a dominant strategy) to the others
  - The problem is that individually rational behavior does not have to be good for the group, which is exactly what the Price of Anarchy measures

Braess' Paradox in Springs



(a) Before (b) After

Figure 2: Strings and springs. Severing a taut string lifts a heavy weight.

Braess' Paradox in Springs

- The same phenomenon appears in physical systems
- A weight hangs on two springs connected in series by a short string; two longer "safety" strings are slack
- In series, each spring carries the full weight, so each of them stretches a lot
- After we cut the short string, the springs hang in parallel (each attached through one safety string) and each of them carries only half of the weight
- Each spring then stretches much less, and the weight *rises*, even though we removed a connection
- The short string plays the role of the new road in Braess' paradox – removing it improves the outcome

Tragedy of the Commons

- Shared resource; getting more increases your reward, but puts strain on the resource





Tragedy of the Commons

	`extract`	`preserve`
`extract`	(50, 50)	(120, 20)
`preserve`	(20, 120)	(100, 100)

Table: Commonize Costs - Privatize Profits Game

- `extract` strictly dominates `preserve` – it yields more no matter what the other player does ($50 > 20$ and $120 > 100$)
- Hence, (`extract`, `extract`) is the unique Nash equilibrium
- Yet both players would be better off if they both preserved the resource (100 instead of 50)
- Each player privatizes the profit of extracting, while the cost of the depleted resource is shared by everyone – the same structure as in the Prisoner's Dilemma



Evaluating the Quality of Strategy Profiles

Motivation

How can we evaluate the quality of an agent, i.e. how well it plays a game?

- In single-agent environments we can easily evaluate how good an agent is based on the score it achieves (e.g. in Atari)
- However, the agent's performance in multi-agent environments directly depends on the quality of its opponents
- ELO-based comparisons are problematic due to intransitivity
- We will use measures that tell us how "close" to an optimal policy we are in terms of performance rather than distance (e.g. KL divergence)

Metrics

We define the following metrics:

- the incentive of player i to deviate

$$\delta_i(\pi) = \max_{\pi_i'} u_i(\pi_i', \pi_{-i}) - u_i(\pi) = u_i(b(\pi_{-i}), \pi_{-i}) - u_i(\pi)$$

- the total incentive to deviate

$$NashConv(\pi) = \sum_{i \in \mathcal{N}} \delta_i(\pi)$$

- the average incentive to deviate

$$Exploitability(\pi) = \frac{NashConv(\pi)}{|\mathcal{N}|}$$

Metrics in Zero-Sum Games

- In zero-sum games, the definition of $NashConv(\pi)$ simplifies to

$$NashConv(\pi) = \sum_{i \in \mathcal{N}} u_i(b(\pi_{-i}), \pi_{-i})$$

- In two-player zero-sum games ($u = u_1 = -u_2$), $NashConv(\pi)$ can be written as

$$NashConv(\pi) = \max_{\pi'_1} u(\pi'_1, \pi_2) - \min_{\pi'_2} u(\pi_1, \pi'_2)$$

Metrics in Zero-Sum Games: Derivation

- In zero-sum games, $\sum_{i \in \mathcal{N}} u_i(\pi) = 0$ for every strategy profile π, so the second term vanishes

$$NashConv(\pi) = \sum_{i \in \mathcal{N}} \big(u_i(b(\pi_{-i}), \pi_{-i}) - u_i(\pi) \big) = \sum_{i \in \mathcal{N}} u_i(b(\pi_{-i}), \pi_{-i}) - \underbrace{\sum_{i \in \mathcal{N}} u_i(\pi)}_{=0}$$

- With two players, $u = u_1 = -u_2$, and $u_i(b(\pi_{-i}), \pi_{-i}) = \max_{\pi_i'} u_i(\pi_i', \pi_{-i})$, we get

$$\begin{aligned} NashConv(\pi) &= \max_{\pi_1'} u_1(\pi_1', \pi_2) + \max_{\pi_2'} u_2(\pi_1, \pi_2') \\ &= \max_{\pi_1'} u(\pi_1', \pi_2) + \max_{\pi_2'} \big(- u(\pi_1, \pi_2') \big) \\ &= \max_{\pi_1'} u(\pi_1', \pi_2) - \min_{\pi_2'} u(\pi_1, \pi_2') \end{aligned}$$

- The last step uses $\max_x(-f(x)) = -\min_x f(x)$

ϵ-Nash Equilibrium

Definition: ϵ-Nash Equilibrium

Strategy profile (π_i, π_{-i}) is an ϵ-**Nash equilibrium** if none of the players can improve by more than ϵ by unilaterally deviating from their policy. Mathematically,

$$\forall i \in \mathcal{N}, \forall \pi_i' \in \Pi_i : \; u_i(\pi_i, \pi_{-i}) \geq u_i(\pi_i', \pi_{-i}) - \epsilon$$

- Iterative algorithms may not be able to find an exact Nash equilibrium due to finite computation time and/or numerical instabilities
- Therefore, we need to use a looser definition of an optimal strategy profile
- If a strategy profile π is an ϵ-Nash equilibrium, it holds that $\max_{i \in \mathcal{N}} \delta_i(\pi) \leq \epsilon$
- Consequently, a strategy profile is a Nash equilibrium $\Leftrightarrow Exploitability(\pi) = 0$


Fictitious Play

Best-Response Dynamics

- The simplest learning dynamics: in **best-response dynamics**, both players simultaneously play a pure best response to the opponent's action from the **previous** round
- Consider the following zero-sum game (payoffs of the row player) starting from (X, A)

	A	B
X	2	-1
Y	-1	1

Time	Play
0	(X, A)
1	(X, B)
2	(Y, B)
3	(Y, A)
4	(X, A)

- The play cycles forever and never converges, as the game has no pure Nash equilibrium

Best-Response Dynamics

- Each action is played in exactly half of the rounds of the cycle, so the average strategies are $\bar{\pi}_1 = \bar{\pi}_2 = (0.5, 0.5)$ with the expected payoff of

$$u(\bar{\pi}) = \tfrac{1}{4}(2 - 1 - 1 + 1) = 0.25$$

- However, the unique Nash equilibrium is $\pi_1^* = \pi_2^* = (0.4, 0.6)$ with the game value of

$$u(\pi^*) = 0.4 \cdot 2 + 0.6 \cdot (-1) = 0.2$$

- The average strategies are not a Nash equilibrium
  - the row player can improve from 0.25 to 0.5 by playing `X`
  - the column player can improve from -0.25 to 0 by playing `B`
  - hence, $NashConv(\bar{\pi}) = 0.5 - 0 = 0.5$
- Best-response dynamics do not converge to a Nash equilibrium, not even on average

Fictitious Play

- Best-response dynamics react only to the opponent's last action, which makes them cycle
- **Fictitious Play** fixes this by best-responding to the opponent's whole history instead
- Each player assumes that the opponent plays a stationary mixed strategy and estimates it by the **empirical distribution** of the opponent's past actions
- In every round, both players **simultaneously** play a **pure** best response to this belief
- It is therefore **myopic**: players maximize their current expected payoff given their current belief, without anticipating future consequences or learning the opponent's update rule
- If multiple best responses exist, any of them can be chosen, so the process is not deterministic

Fictitious Play

- The belief about player i is exactly player i's **average strategy**

$$\bar{\pi}_i^t = \frac{1}{t} \sum_{k=1}^{t} \pi_i^k,$$

  where π_i^k is the (pure) strategy played by player i in round k
- A single iteration of the algorithm then updates the average strategies as

$$\bar{\pi}_i^{t+1} \in \left(1 - \frac{1}{t+1}\right) \bar{\pi}_i^t + \frac{1}{t+1} b(\bar{\pi}_{-i}^t)$$

- In other words, every player moves their average strategy a small step towards a best response to the opponent's average strategy

Fictitious Play Example

Let's consider the Matching Pennies game and simulate a couple of steps of Fictitious play. Let n_i^t be the running count of the number of times player i played each action.

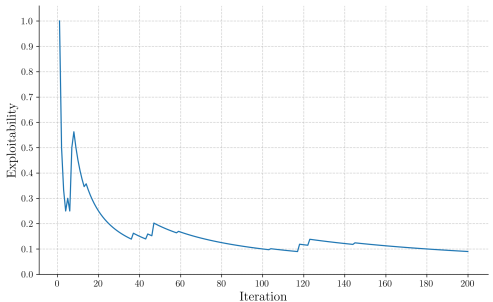
	heads	tails
heads	(1, -1)	(-1, 1)
tails	(-1, 1)	(1, -1)

Table: Matching Pennies

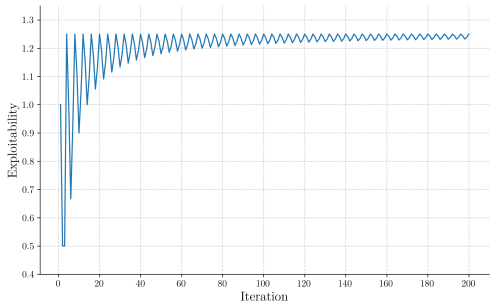
Time	n_1^t	n_2^t	Play
0	(0, 0)	(0, 0)	(h, h)
1	(1, 0)	(1, 0)	(h, t)
2	(2, 0)	(1, 1)	(h, t)
3	(3, 0)	(1, 2)	(t, t)
4	(3, 1)	(1, 3)	(t, t)
5	(3, 2)	(1, 4)	(t, t)
6	(3, 3)	(1, 5)	(t, h)
7	(3, 4)	(2, 5)	(t, h)
8	...	...	...

Fictitious Play vs. Best-Response Dynamics

- In Fictitious Play, the **average** strategies converge to a Nash equilibrium in certain classes of games (**average-iterate** convergence), even though the actions actually played do not converge in general; best-response dynamics may cycle forever instead



(a) Fictitious Play



(b) Best-response dynamics

Convergence of Fictitious Play to Pure Strategies

- Let us now study the asymptotic behavior of the sequence of strategy profiles $\{\pi^t\}$ produced by FP, i.e. the convergence properties of the sequence $\{\pi^t\}$ as $t \to \infty$
- We say the sequence $\{\pi^t\}$ settles to π^*, if there exists T, s.t. $\forall t \geq T : \pi^t = \pi^*$
- The following theorem formalizes the property that if the sequence $\{\pi^t\}$ converges, then it has to converge to a Nash equilibrium of the game

Theorem

If the sequence $\{\pi^t\}$ converges to π^*, then π^* is a pure strategy Nash equilibrium. Moreover, suppose that for some t, $\pi^t = \pi^*$, where π^* is a **strict** Nash equilibrium. Then $\pi^{t'} = \pi^*$ for all $t' \geq t$.

Convergence of Fictitious Play to Mixed Strategy

- The sequence $\{\pi^t\}$ converges to a mixed strategy profile π^* in **the time-average sense**, if for each player $i \in \mathcal{N}$ and for all actions $a_i \in \mathcal{A}_i$, we have:

$$\lim_{T\to\infty} \frac{\sum_t \mathbb{1}(\pi_i^t = a_i)}{T} = \pi^*(a_i)$$

Theorem

If the sequence $\{\pi^t\}$ converges to π^* in the time-average sense, then π^* is a mixed strategy Nash equilibrium.

Convergence of Fictitious Play

Games in which Fictitious Play converges are said to have **the fictitious-play property**. The algorithm has been proven to converge for the following classes of games:

- two-player zero-sum games
- two-player non-zero-sum game, where each player has at most two strategies
- games solvable by Iterated removal of strictly dominated strategies
- identical interest games; games where all players have the same payoff function

Convergence of Fictitious Play

On the other hand, in games such as the **Shapley game**, Fictitious Play can cycle indefinitely and fail to converge, depending on the initial conditions.

	x	y	z
a	(2, 1)	(0, 0)	(1, 2)
b	(1, 2)	(2, 1)	(0, 0)
c	(0, 0)	(1, 2)	(2, 1)

Table: The Shapley Game

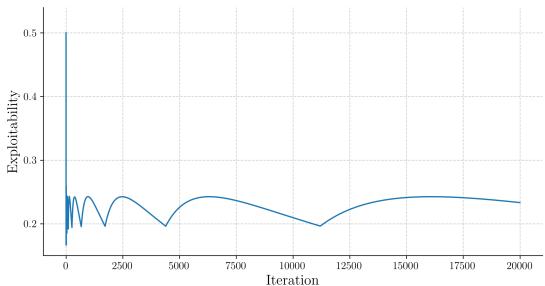


Figure: Convergence of FP when starting with (a, x)

Week 3 Homework

You can find more detailed descriptions of homework tasks in the GitHub repository.

1. Strategy profile evaluation ($\delta_i(\pi), NashConv(\pi), Exploitability(\pi)$)
2. Fictitious play
3. Exploitability convergence plots